Airway, Drug Assisted

AR-03 | Airway & Respiratory

Quick flow

01 DAI Indicated + Authorized? [Paramedic]

Airway protection failure • oxygenation/ventilation failure • impending compromise; not for convenience.

02 Prepare Completely [Paramedic]

Two Paramedics • monitor • suction • BVM/PEEP • primary and rescue airway • surgical-airway plan.

03 Preoxygenate + Resuscitate [Paramedic]

High-flow oxygen, NIPPV or BVM/PEEP; correct hypoxia and hypotension before paralysis.

04 Induction + Paralysis [Paramedic]

Select one induction agent, use the appropriate paralytic, and proceed immediately to airway placement.

05 Airway Successful? [Paramedic]

Waveform $EtCO_2$, bilateral exam, tube depth, SpO_2, and securement. If not: BVM/BIAD and AR-02.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Post-Airway Management [Paramedic]

Pain-first analgesia/sedation • ventilate 10-12/min • continuous monitoring • transport.

Overview

- Drug-assisted airway (DAI) is an optional, high-risk procedure requiring specific agency authorization, documented competency, performance improvement, and at least two Paramedics on scene.
- Correct hypoxia and hypotension before administering a sedative-paralytic sequence whenever possible; resuscitation takes priority over intubation.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Assist with preoxygenation, suction, BVM, equipment preparation, physiologic monitoring, and backup-airway readiness.
- Continue oxygenation and ventilation throughout preparation and between attempts.

AEMT

- Perform all EMT care plus authorized vascular access, rescue BIAD support, monitoring, and assistance with the failed-airway plan.

Paramedic

- Confirm indications, authorization, two-Paramedic staffing, monitoring, preoxygenation, suction, primary device, rescue BIAD, surgical-airway equipment, and a failed-airway plan before medication.
- Obtain two IV/IO access points when feasible, but do not delay life-saving oxygenation or ventilation for a second access attempt.
- Correct hypoxia and hypotension before paralytic administration, limit laryngoscopy attempts, verify with waveform capnography, and begin pain-first post-intubation care immediately.

Indications

- Failure to protect the airway, inability to oxygenate, inability to ventilate, or impending airway compromise when less-invasive care is inadequate and DAI is authorized.

Treatment steps

- Do not perform DAI solely for convenience, because BVM is effective, or for a short transport when BVM or BIAD safely maintains oxygenation and ventilation.
- Preoxygenate with high-flow oxygen, NIPPV when appropriate, or two-person BVM with PEEP. Prepare suction, an appropriately sized primary airway device, BIAD, and surgical-airway rescue before medication.
- Use the AM-05 or AM-07 resuscitation pathway for peri-intubation shock. Treat hypotension before paralysis; use norepinephrine when indicated and open the dedicated Push-Dose Epinephrine reference only for its approved temporary-bridge use.
- Select one induction medication, then give the appropriate paralytic and proceed immediately to airway placement. Do not routinely repeat rocuronium before post-intubation analgesia and sedation are established.
- Limit laryngoscopy to three total attempts. Each subsequent attempt must change the provider, approach, positioning, or equipment; move immediately to AR-02 when oxygenation or ventilation is ineffective.
- Confirm every advanced airway with continuous waveform $EtCO_2$, bilateral chest assessment, SpO_2, tube depth, and securement. Reassess after each movement.
- After airway placement, treat pain first, then sedation. Ventilate an adult at approximately 10-12 breaths/min with visible chest rise and use $EtCO_2$ and clinical response to avoid hyperventilation.
- Document indication, authorization, staffing, preoxygenation, medications and doses, oxygenation/hemodynamics before and after medication, attempt count, airway confirmation, ventilatory support, sedation/analgesia, and required quality-improvement form.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Medications

Ketamine

1-2 mg/kg IV/IO

- Select one induction agent. Ketamine is generally preferred when hypotension or bronchospasm is present.
- If vascular access is unavailable: 4 mg/kg IM, maximum 400 mg; establish IV/IO access and proceed with airway management.
- Pediatric use requires a direct online order from the Claiborne EMS Medical Director or Assistant Medical Director.

Etomidate

0.3 mg/kg IV/IO

- Select one induction agent when hemodynamic response is uncertain.

Succinylcholine

1-2 mg/kg IV for RSI

Rocuronium

1 mg/kg IV/IO when succinylcholine is contraindicated or a longer duration is needed

- Do not routinely repeat before post-intubation analgesia and sedation are established.

Fentanyl

25-50 mcg IV/IO; repeat cautiously as needed, maximum 100 mcg

- Use lower or fractionated doses with shock, older age, or other hemodynamic risk.

Midazolam

2-5 mg IV/IO; repeat cautiously as needed

- Use after analgesia when sedation is needed.

Warnings / precautions

- This procedure removes protective airway reflexes and spontaneous breathing; do not begin without complete rescue capability.
- At least two Paramedics must be present under the approved protocol.
- DAI during unresolved hypoxia or hypotension substantially increases cardiac-arrest risk.
- No more than three total laryngoscopy attempts. Effective BVM is an acceptable definitive field airway.
- A paralyzed patient may be awake and in pain without visible movement; establish analgesia and sedation promptly after airway placement.
- Complete the required airway evaluation and performance-improvement documentation.

Clinical pearls

- When BVM maintains saturation at least 90%, continued basic airway management may be safer than DAI during a short transport.
- BIAD is preferred while resuscitating severe hypoxia or hypotension.
- Norepinephrine may be used for peri-intubation hypotension; follow AM-05 and the dedicated Push-Dose Epinephrine reference for its temporary-bridge safeguards.

Related pathways / contacts

AM-05 Hypotension / Shock - Open for peri-intubation hypotension or shock.

AM-07 Crashing Medical Patient - Open for immediate physiologic resuscitation when the patient is critically unstable.

AR-02 Adult Failed Airway - Open immediately when oxygenation or ventilation is ineffective or further laryngoscopy is unsafe.

AR-08 Post-Intubation / BIAD Management - Open for continued analgesia, sedation, ventilation, and monitoring after airway placement.

Push-Dose Epinephrine - Medication Reference - Open the approved labeled preparation, dosing, and two-bolus limit.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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